

An Integrated UAV Navigation System Based on Geo-registered 3D Point Cloud

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Abstract—The autonomous navigation of unmanned aerial vehicles (UAVs) require a lot of sensing modalities to improve their cruise efficiency. This paper presents a system for autonomous navigation and path planning of UAVs in GPS-denied environment based on the fusion of geo-registered 3D point clouds with proprioceptive sensors (IMU, odometry and barometer) and the 2D Google maps. The contributions of this paper are illustrated as follows: 1) combination of 2D map and geo-registered 3D point clouds; 2) registration of local point cloud to global geo-registered 3D point clouds; 3) integration of visual odometry, IMU, GPS and barometer. Experiment and simulation results demonstrate the efficacy and robustness of the proposed system.

I. INTRODUCTION

In the recent years, quadrotor UAVs have drawn a lot of attentions from both military and urban civil applications due to their small size, low cost, hovering capability, and high maneuverability [1]. Those civil applications include surveillance and monitoring, disaster and emergency management, precise agriculture, power line and pipeline inspection, search and rescue, aerial photography, etc. Those military applications include supply transportation, reconnaissance and surveillance, border control, electronic warfare, etc [2].

Reliable navigation and path planning are keys to autonomous UAV technologies, which rely on the accurate self-location techniques through using GPS signals, IMUs and barometers. However, the GPS signals can be interfered for various reasons, such as extreme atmospheric conditions, obstacles and terrains, television broadcast, and inconvenient positions of satellites [3].

And IMUs are prone to generate systematic errors: when GPS signals are unavailable or corrupted, the estimate of UAV states(position, velocity and attitude) provided by IMUs alone will drift quickly and be unreliable after a few seconds [4]. Besides, when a UAV is close to the ground, neither GPS nor IMU nor pressure based barometer can provide accurate measurements of its vertical position and movement. All of the above finally cause errors in UAV self-localization results.

For path planning of UAVs, especially in large area, it is an imperative task about seeking an optimal or near-optimal flight path between a starting point and the desired destination point considering the different kinds of constraints under complex dynamic field environment. Many methods have been proposed for UAV path planning, such as artificial potential field (APF) method, evolutionary computation, A*

algorithm, genetic algorithm and so on [5]. However, those methods can hardly solve the contradiction between the global optimization and excessive information [6].

Autonomous UAVs require reliable path planning and navigation. However, those technical challenges should be overcome:

- 1) accurate and reliable navigation data (position, velocity, attitude and altitude) are needed for guidance and control of UAVs;
- 2) navigation and path-planning algorithms must be computationally efficient as the flight environments could be constantly changing;

Currently, video cameras become a standard sensor for UAVs for their reduced sizes and costs but increased performance. And vision-based approaches to enhancing UAV navigation and path planning are attracting increasing research interests. The images taken by onboard-cameras can provide a large amount of information which can be utilized to solve navigation and path planning related problems. Since GPS, IMU, barometer and onboard-cameras have different measurement strengths and limits, they can complement each other in different situations and provide accurate and real-time navigation data for UAVs.

In this paper, we propose a framework of real-time UAV based autonomous navigation and path planning. Compared with most navigation and path planning methods described above, we develop a more complete and robust system with both semantic geo-registered cloud points database, neural network based object database, and address the mentioned challenges accordingly.

The contributions of this paper include

- 1) Design an autonomous navigation and path planning system for UAVs;
- 2) Combine geo-registered 3D point clouds database and 2D digital maps for vision-based UAVs navigation;
- 3) Fusion of vision-based navigation information with proprioceptive sensors (IMU, odometry and barometer) to provide accurate and reliable navigation data for UAVs;
- 4) Utilize a neural network based object detection method to realize dynamic obstacles avoidance for UAVs;
- 5) Utilize semantic cloud points to realize static obstacles avoidance for UAVs;

This paper is organized as follows. Section II introduces the related work on vision-based UAV navigation and path planning. Section III describes the whole system architecture and states problems. Section IV presents the integrated

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navigation and path planning system framework, including georegistration of point clouds using geographic data, registration of point clouds captured by on-board camera to geo-registered 3D point clouds database, neural network based obstacles avoidance and semantic point clouds based obstacles avoidance, and fusion of vision-based navigation information with proprioceptive sensors. Section V provides the results and discussions. Section VI concludes the paper and outlines future work.

II. RELATED WORK

A. Vision-based UAV Navigation

Outdoor UAV navigation usually involves obstacles-avoidance, landmark detection, map building and updating, and attitude estimation. And vision-based navigation technology plays an important role in UAV outdoor navigation. It is typically approached by matching the aerial image sequence with geo-registered image database [7]. Researchers have developed several vision-based methods to provide information for UAV navigation. These methods are mainly categorized into scene matching based, Kalman filter based [8], simultaneous localization and mapping (SLAM) based [8], 3D point cloud registration based method. For scene matching based navigation method, natural landmarks [9] or man-made markers [10], [11] are utilized to improve scene matching efficiency and accuracy. However, the performance of these methods were greatly depend on number, features and distribution of landmarks.

A relatively early work on Kalman filter based vision-based UAV technique is [12]. In this estimator, the position information is obtained from an extended Kalman filter, with the UAVs position and velocity as its states and the inter-frame displacement measurements as observation data. However, the estimator did not provide experimental and simulation results. Another Kalman filter based technique was proposed in [13], [14] that does take rotation into consideration. It estimates the airplanes position by estimating its translation and rotation using optical flow and other sensory data (e.g., altitude information). Sometimes it combines the vision odometry information and IMU measurements to estimate UAV's position and attitude [15], [16]. The difficulties of the methods using fusion information of stereo vision and IMU were easily interfered by complicated environment and easily lead to drifting errors. Numerous research works were carried out to apply the SLAM technique to UAV [17], [18].

SLAM which was originally developed for estimating ground robot's position and a set of landmarks' position, it has been adopted to vision-based UAV navigation. For instance, the approach in [19] requires man-made ground landmarks with known sizes and that in [20] requires the known 3D coordinates of some ground landmarks. In some applications, landmarks cannot be planted ahead of time, not to mention specific information of landmarks. Therefore, SLAM is still a challenge when applied to large areas. Continuous generation of a 3D point cloud for underground vehicle navigation in dynamic and unknown environments were described in [21], [22] etc. However, this method has

the limitations of the rapid and cost effective capturing of 3D data from larger outdoor environment.

B. Path Planning for UAV

In recent years, there are many methods that produce either 2-D, or 3D trajectories for guiding UAV in different kinds of environments, such artificial potential field (APF) method, genetic algorithm and A* algorithm [23], [24], [25], [26]. Due to its advantages of simple algorithm structure, the concise mathematical description and the convenience for real-time control, the APF method is commonly used as a path planning and obstacle avoidance candidate technique [27]. Furthermore, it also can be utilized to solve the collision avoidance problem of the multi-UAV system by changing the source of the artificial potential field.

Based on the APF, a solution for collision and obstacle free formation flight and reconfiguration of groups of autonomous helicopters was presented [28]. Similarly, the APF method is applied to solve the problem of collision and terrain avoidance for multiple UAVs, tracking a low-altitude trajectory [29]. The genetic algorithms have also been used for solving the UAV path-finding problem in 3D environment. Although these optimization methods can effectively solve the problem of path planning, they have existed premature convergence phenomenon. Therefore, the local minimum problem is APF's biggest problem.

For A* algorithm, it calculates the cost of the current nodes to reach other nodes and uses the heuristic information to evaluate the node, based on that then choosing the lowest total cost node as its expanded node and finally get the optimal path [30]. A* algorithm is commonly used in the engineering algorithm; not only has it a higher searching efficiency, but also guarantees the optimal path. Therefore, in this paper, A* algorithm is utilized to plan path for UAV navigation.

III. SYSTEM SETUP

A. System Diagram

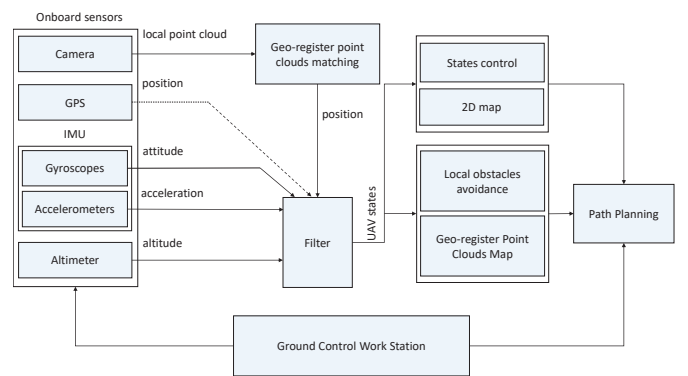


Fig. 1. System diagram.

Fig. 1 shows our system diagram. Our system includes five components: 1) onboard sensors module, 2) data processing module, 3) system control module, 4) path planning module,

5) ground control work station. The state vector of UAV includes real-time imaging, altitude, azimuth and coordinates of the world.

These state vectors can be obtained from the onboard sensors which contain camera(s), GPS, IMU, and altimeter. And gyroscopes and accelerometers comprise the IMU sensors. We use the images from on-board camera to reconstruct 3D point clouds and register the GPS coordinates into the 3D point clouds. During the procedure of UAV flight, there are many noises to have an effect on the data obtained from these onboard sensors. The other component of data processing module is a filter to make sure the data without noises pollution.

The control module includes states control with 2D map, local obstacles avoidance with geo-register point clouds map. The function of this module is to realize the navigation, localization and obstacles avoidance. The path planning module is used for find the most optimized path for flying when the UAV is in the auto-pilot mode.

The ground control work station is integrated with the GIS database to determine the global coordinates of the target and provide the target trajectory prediction using geographical information such as the high building. And the user can interact with the UAV through the ground command center such as select the target from the video monitor or change the UAV flight course.

B. Dynamic model and Algorithm overview

When GPS is lost, the calibrated IMU data and baro-altimeter data is utilized to provide UAV states. For INS/baro-altimeter data association system with discrete observations, we should provide a dynamic model to suppress the errors drifting with time. At time k , the estimated UAV state vector is as the following:

$$\mathbf{x}_k^w = (\mathbf{v}^w, \mathbf{g}^w, \mathbf{h}^w, \mathbf{a}_v^w), \quad (1)$$

where $\mathbf{x}_k^w$ is an estimated vector of the time k ; $\mathbf{v}^w$ represents the velocity of the UAV; $\mathbf{g}^w$ represents the gyroscopes of UAV; $\mathbf{h}^w$ represents the height vector of UAV; $\mathbf{a}_v^w$ represents the acceleration of IMU mounted on UAV. The overall procedure of our integrated UAV navigation system is presented in table I.

IV. NAVIGATION AND PATH PLANNING

A. Integrated GPS/Vision-based Navigation System

Depending on the availability of GPS information, the integrated navigation system works with two models. When GPS measurements are available, a sixteen-state Kalman filter estimates the errors of a strap down calculation which are corrected subsequently, assuring hereby the long-term accuracy of the navigation solution. The state vector of this filter comprises three errors in position, three errors in velocity, three attitude errors, six errors of the inertial sensor bias estimates, as well as the error of the baro-altimeter bias estimate. Additionally, the accelerometer and gyroscope biases are estimated and corrected to improve the accuracy of UAV states.

TABLE I
VISUAL OBJECT TRACKING ALGORITHM

Algorithm 1 Integrated UAV Navigation Algorithm	
Input:	GPS signal $\mathcal{T}_i (i = 0, 1, 2, \dots)$
Output:	GPS signal $\mathcal{T}_i^c (i = 0, 1, 2, \dots)$, GPS signal set $\mathcal{T}$
1:	repeat
2:	Get SNR of the current GPS snr ;
3:	if $(30 \leq snr \leq 50)$ then
4:	$\mathcal{T}_i^c = \mathcal{T}_i$;
5:	Periodically Update and correct the signal of IMU, altimeter;
6:	else
7:	Get the altitude (h), velocity (v), orientation information d ;
8:	EKF $\leftarrow \{h, v, d\}$ and get GPS $\mathcal{T}_i^c$;
9:	end
10:	$\mathcal{T} \leftarrow \mathcal{T}_i$
11:	Update model $\mathcal{T}$;
12:	Obstacle avoidance;
13:	until end of flight

When GPS is lost, the vision-based navigation model is selected. The positions of UAV are calculated by registering the local point cloud or aerial images to hybrid database. After an appropriate initialization of the state and the covariance matrix using data provided by the sixteen-state Kalman filter, the calibrated IMU data and baro-altimeter data is utilized to provide UAV states.

$$\begin{cases} X(k) = \emptyset_{k,k-1}X(k-1) + B_kU_k + \Gamma_kW(k) \\ Z(k) = C_kX(k) + \nu(k) \end{cases} \quad (2)$$

where $X(k)$ is an estimated vector of the time k ; $\emptyset_{k,k-1}$ is the state transition matrix from time $k-1$ to time k ; $x(k-1)$ is a vector of the time $k-1$; Γ_k is the system noise matrix corresponding to the system noise $w(k)$ in the time k ; B_k is the control-input matrix which is applied to the control vector U_k ; $Z(k)$ is an observation (or measurement) of the true state $X(k)$ at time k ; C_k is the observation matrix which maps the true state space into the observed space and $\nu(k)$ is the observation noise.

We don't make sure that the GPS module is always in work. There are many factors including being in tunnel or forest that can reduce the positioning accuracy of GPS module. When the entropy of GPS module is higher than the threshold, the state vector $X(k)$ includes the information of GPS, IMU and baro-altimeter. When the entropy of GPS module is lower than that of the threshold, the information obtained based vision will replace the position of the GPS module.

B. Visual Inertial Odometry

To improve the precision of the vision-based navigation in GPS-denied areas, we utilize visual inertial odometry with filter-based to realize navigation and localization. We add both aerial image features and IMU data into the state vector at the time k . The state vector at time k includes the pose of UAV, scale factor, current attitude errors, gyroscope biases and a group of image features. The brute force feature matching computation cost of these features are $O(M^2)$, and M is the dimension of these features. It will reduce the performance of real-time navigation to use all of features of

the current aerial image. Then we just only extract the fixed ratio of total number of features to improve the performance of real-time navigation.

In this section, we utilize the BRISK [31] algorithm to extract the aerial image features. To realize the registration between current frame and its last key frames, we utilize the Brute-force descriptor matcher. Then we can construct the reprojection error of these frames with bundle adjustment to find the mapping relationship between 2D and 3D. With these prior knowledge and state vector of previous frame, we can use the EKF to estimate the current state vectors. There is error term of IMU which represents the difference between estimated state vector and parametric variable. Combine these two error vectors, we can derive the following cost function:

$$J(x) := \sum_{i=1}^I \sum_{k=1}^K \sum_{j \in \mathcal{J}(i,k)} e_r^{i,j,kT} W_r^{i,j,k} e_r^{i,j,k} + \sum_{i=1}^{K-1} e_s^{iT} W_s^k e_s^k, \quad (3)$$

where i is the i th frame k is the index of the i th frame of camera; j is the index of image features; $W_r^{i,j,k}$ represents the information matrix; W_s^k represents the error of information matrix.

The 3D point clouds which are reconstructed with our aerial images are registered with global-position-system(GPS) coordinates. We can utilize the geo-registered 3D point clouds to rectify the errors of IMU and reprojection.

C. Semantized point clouds for UAV Scene Understanding

Point cloud is an important type of geometric data structure. Compared to 2D image, 3D point cloud could provide more geometric information for UAV to better percept and understand the surrounding environment. Fig. 2 illustrates the construction procedure of constructed 3D semantized point clouds. As shown in the Fig. 2, we registered 3D point clouds with the GPS coordinates corresponding to the known 2D map, like Google Map, or Bing map etc..

As we all known, the 2D map always contain the property of objects. Then we can utilize the known geometry information of 2D map to label the most of objects in the 3D point clouds. It can reduce the computation cost of semantic segmentation and labeling by means of geo-registered 3D point clouds and 2D map. We use the state-of-art object detection neural network [32] to detect these objects which are used for the prior knowledge for the condition-random-field (CRF) algorithm to accurately segment these unlabeled objects for semantic labelling.

Now we can construct the octree map to encode the property of each voxel. Using octree map has merits [33] on 1) supporting multi-resolution map queries; 2) storing occupancy probability in nodes; 3) compacting in memory and on disk and efficient access and queries; 4) flexible extension of mapped area; 5) lossless compression. The octree map can help UAV to cognitive its complex surroundings. Especially when UAV encounter some obstacles, like building, power lines and tress, in the denied GIS area, the UAV can efficiently avoid the collision with obstacles.

As to to automatically recognize object during the process of operating flight and prepare for the keeping UAV away from obstacles, we extract image features and achieve the classification through CNNs.

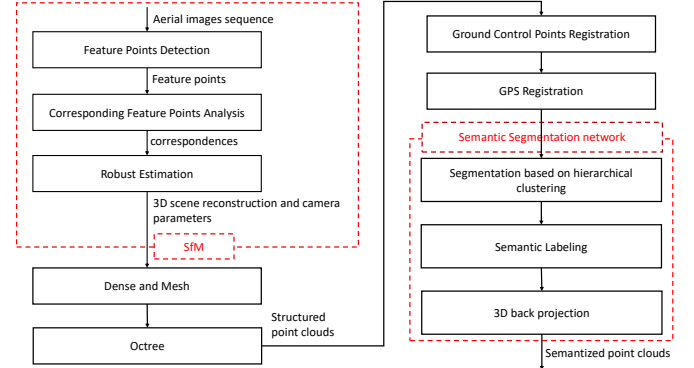


Fig. 2. (left) The aerial images sequences captured by on-board camera are applied to generate point clouds of surrounding environment with using GlobalACSfM[34] algorithm. Octree Algorithm is applied to structuralize 3D point clouds to improve data access and queries efficiency; (right) Semantic3D.net algorithm[35] is utilized to semantic segmentation point clouds. Ground control point registration[36] is performed to restore the semantized point clouds scene to known 3D coordinates. Geodesy data(i.e GPS or XYZ prior) is registered to SfM camera center positions to limit the number of pair by matching only neighbour images so that speed up the pairwise features matching.

V. EXPERIMENT RESULTS

Fig. 3 shows the experimental setup. The sensors onboard the quadrotor UAV includes differential GPS (DGPS), IMU, CCD camera with large view field, altimeter. The sensor locations are showed in Fig. 3. Integrated aerial zoom camera based on DJIs class leading Zenmuse X3 was installed on the UAV, its large focal length range from 22mm to 77mm makes it ideal for aerial photography. The sensors' related

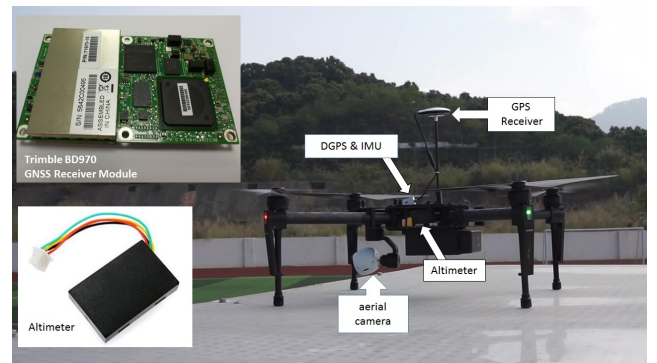


Fig. 3. Experiment setup

parameters are listed in Table II.

Fig. 4(a) shows some different illustrations of our 3D campus. These point clouds are reconstructed with aerial images by camera mounted on the UAV. Fig. 4(b) shows Octree map which is a hierarchical tree data structure from point cloud data (Fig. 4(a)). We utilize the jet color to illustrate different altitude of 3D points of our 3D campus. The Octree

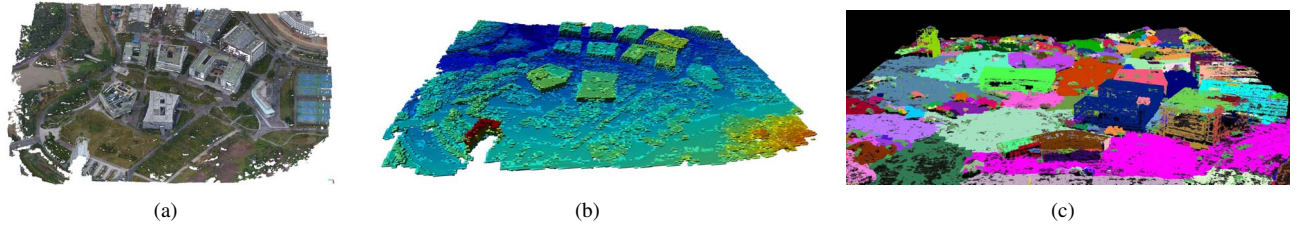


Fig. 4. Illustrations of 3D point clouds.(a) Image-based point clouds; (b) Octree map generated from (a), showing occupied voxels only; (c) Semantic segmentation of (a).

TABLE II
PERFORMANCE PARAMETERS FOR SENSORS OF INTEGRATED
NAVIGATION SYSTEM

Sensors	Accuracy	Parameters	Max Rate
DGPS	Horizontal: $\pm(8\text{mm}+1\text{ppm})$ Vertical: $\pm(15\text{mm}+1\text{ppm})$	N, E, Alt(m); $V_{xy}, V_z(\text{m/s})$	50Hz
IMU	Gyro scale: $<2\%$ Gyro bias(deg/hr): 20 Acce scale: $<1\%$	p q r(deg/s) $a_x a_y a_z$	50Hz
Altim	detect range: 40m accuracy 0.02m	height	

is a hierarchical data structure for spatial subdivision in 3D space. Accordingly, we can set different resolution of Octree to adjust the size of 3D point clouds model correspondingly. The resolution of Octree is an termination condition to decide whether the cubic volume is the recursively subdivided into eight sub-volumes. There are many small cubic in Fig.4(b). Each small cubic represents a node in a Octree, usually called a voxel. The Octree can improve the efficiency of Computation and storage. Fig. 4(c) shows an semantic labeling of 3D point clouds. An semantic labeling of 3D point clouds can help a robot, unmanned vehicle(UV) and UAV to understand the nature of its surroundings. As we can see from Fig. 4(c), the structured object model, like buildings and roads, can be easily segmented from the point clouds. On the other hand, these unstructured objects model, like grass and river, cannot be accurately segmented from the our 3D campus model. Due the flight of UAV in the open air, these unstructured objects models don't have an effect on the flight of UAV.

Fig. 5 shows an example trajectory generated by our UAV with A* algorithm. The curve (red) is trajectory of UAV with A-star path planning. The length of flight trajectory of UAV is about 250m. The triangle (red) and circle (blue) are the start point and destination point of our UAV respectively. In this experiment, the GPS signal is removed to verify our integrated navigation system, and recorded to verify the accuracy of our integrated navigation system. Accordingly, we always maintain the drones flying at an altitude below 40 meters to verify the performance of obstacle avoidance. So we can find the UAV across under the fly-over between two buildings, the height being about 15 meters.

Fig. 6 shows a comparison from GPS receivers with our integrated UAV navigation system. Fig. 6(b) shows the GPS

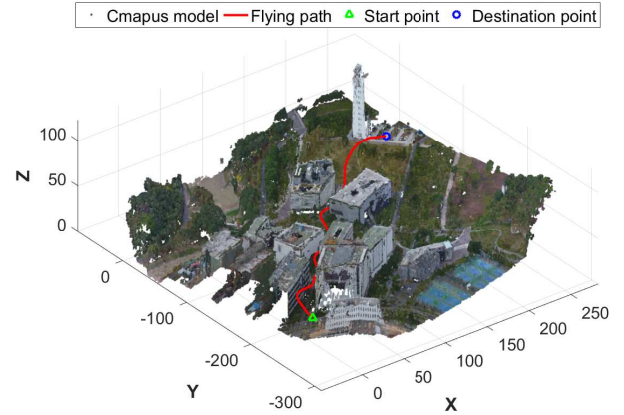


Fig. 5. Example of trajectory generated by our UAV. Our 3D campus model is shown with point clouds; The curve (red) is trajectory of UAV with A* path planning; the triangle is the start point of UAV and the square is the destination point of UAV.

error in the X direction. Fig. 6(c) shows the GPS error in the Y direction. We can find the error range in the X direction is $[-4, 4]$ meters, and in the Y direction is $[-6, 6]$ meters. The range of error is allowable and achieve the general civilian GPS positioning capabilities. The error come from two aspects: 1)accumulation error of IMU and drifting with time, 2) inaccurate vision-based navigation.

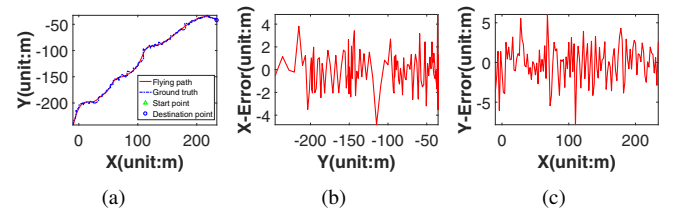


Fig. 6. Overview of GPS signal. (a) GPS comparison between estimated and ground truth; (b) Error in the X direction; (c) Error in the Y direction

VI. CONCLUSIONS

We presented an integrated UAV navigation system for flight path planning during the UAV in GPS-denied area. This system is a coarse-to-fine navigation system. During the coarse phase, this integrated system is based on GEO-registered 3D point clouds and 2D map for vision-based UAV navigation. Fusion of vision-based navigation information with proprioceptive sensors (IMU, odometry and barometer) is to provide accurate and reliable navigation data for UAVs. The deep neural network and semantic point clouds provide

reliable advice for path planning and obstacle avoidance. The Integrated UAV Navigation System Based on Geo-registered 3D Points Cloud performs competitively with respect to the state-of-the-art methods in terms of localization accuracy and navigation.

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